





section 7

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HDPE series



Shurjoint offers a series HDPE couplings and adapters for joining HDPE pipe. The use of HDPE (high density polyethylene) pipe continues to grow in popularity as its benefits over traditional materials are realized in a variety of service applications. The benefits of an HDPE pipe system include a longer service life, increased flexibility, reduced weight, increased resistance to corrosion, chemicals, and fatigue, as well as superior flow characteristics. HDPE is now commonly used in service applications including municipal water and waste water, water distribution and transport, mining, slurry and many other general and industrial applications.

Shurjoint HDPE couplings provide a fast and easy way to mechanically join HDPE pipe. A series of sharply machined teeth securely grip the pipe as the bolts are tightened, resulting in a leak-free joint. The Shurjoint joining method eliminates the need for costly heat fusion equipment. The highly restrained joint allows long lengths of pipe to be pulled from one area to another. With the removal of a few bolts one can easily access the system for cleaning, maintenance, changes and or system expansion.

Shurjoint HDPE couplings are designed to join IPS HDPE pipe, DR32.5 to 7.3, conforming to ASTM D2513, D3350 and/or ANSI/AWWA C901, and ISO HDPE pipe, SDR 9 to 26, conforming to ISO 4427-1/2.

As the ductile iron coupling is much stronger than HDPE pipe itself, pressure ratings of HDPE couplings are determined by the pressure rating of HDPE pipe used. Pressure ratings of HDPE pipe vary depending on DR or SDR (standard dimension ratio) and design stress of the material.

pressure ratings (psi) - IPS size

pipe dimension ratio (DR)	PE4710 PE100	PE3608 PE3408
DR 7.3	317	265
DR 9	250	200
DR 11	200	160
DR 13.5	160	130
DR 17	125	100
DR 21	100	80
DR 26	80	65
DR 32.5	63	50

design stress: PE4710 1000 psi, PE3608 & 3408 800 psi

DR (Pipe Dimension Ratio)
$$DR = \frac{D}{t}$$

where:
D = pipe outside diameter, in
t = pipe minimum wall thickness, in

pressure ratings (bar) - ISO size

pipe dimension ratio (DR)	PE100	PE80
SDR 9	20	16
SDR 11	16	10
SDR 17	10	6.3
SDR 26	6.3	4

design stress: PE100 8.0 MPa, PE80 5.0 MPa

SDR (standard pipe dimension ratio)
$$SDR = \frac{D}{t}$$

where:
D = pipe outside diameter, mm
t = pipe minimum wall thickness, mm

installation instructions

1. marking: Use a marker and measuring tape to place “Gasket Centering Marks” and “Coupling Centering Marks” on each pipe end per the installation instructions.



2. install gasket: Place a gasket over the pipe ends and center the gasket in between the first set gasket centering marks. The pipe ends should always be butted against each other.



3. mount housings: Place the housings over the gasket, ensure the gasket stays centered between the first set marks made on the pipe ends and the housings are properly centered between the second set coupling center marks. Also make sure that housing tongue and groove (T&G) mate correctly.

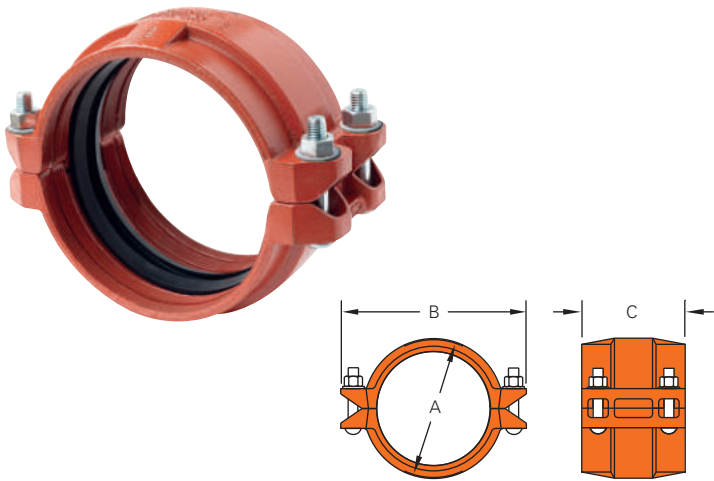


4. tighten nuts: Tighten nuts alternately and equally until the bolt pads meet and make metal-to-metal contact. Repeated alternate tightening will reduce tightening torque considerably. Tighten nuts by another one quarter to one half turn to make sure the bolts and nuts are snug and secure.



caution: refer to the Shurjoint installation instruction manual for complete instructions. Shurjoint HDPE couplings are not intended for use on PVC, PP or other materials. Do not use standard soap based lubricant on HDPE pipe. Shurjoint recommends the use of a silicone based lubricant with the HDPE series. In order to avoid injuries from the sharp machined teeth, wear gloves when handling.

H305 HDPE coupling



The Shurjoint Model H305 HDPE Couplings feature four bolt housings and a series of sharply machined teeth which securely grip the pipe as the bolts and nuts are tightened. The result is a leak-tight joint that is as strong as or stronger than the pipe itself. The H305 also features a contoured housing with integral ramps along the outside diameter to help the coupling slide over most obstacles during the relocation of pipe runs.

dimensions

H305 IPS size



nom. size	pipe O.D.	dimensions			bolt		weight
		A	B	C	no.	size	
2"	2.375"	3.39"	5.24"	4.57"	4	½ x 2¼"	5.7 lbs
50	(60.3 mm)	(86 mm)	(133 mm)	(116 mm)			(2.6 Kgs)
3"	3.500"	4.61"	6.50"	4.57"	4	½ x 3"	7.9 lbs
80	(88.9 mm)	(117 mm)	(165 mm)	(116 mm)			(3.6 Kgs)
4"	4.500"	5.75"	7.95"	5.75"	4	½ x 3"	11.4 lbs
100	(114.3 mm)	(146 mm)	(202 mm)	(146 mm)			(5.2 Kgs)
5"	5.563"	6.66"	8.25"	4.63"	4	½ x 3"	9.9 lbs
125	(141.3 mm)	(169 mm)	(210 mm)	(118 mm)			(4.5 Kgs)
6"	6.625"	7.87"	10.75"	5.87"	4	¾ x 3½"	18.0 lbs
150	(168.3 mm)	(200 mm)	(273 mm)	(149 mm)			(8.2 Kgs)
8"	8.625"	10.39"	13.11"	6.02"	4	¾ x 3½"	28.4 lbs
200	(219.1 mm)	(264 mm)	(333 mm)	(153 mm)			(12.9 Kgs)
10"	10.750"	12.52"	15.71"	6.50"	4	¾ x 4¾"	43.6 lbs
250	(273.0 mm)	(318 mm)	(399 mm)	(165 mm)			(19.8 Kgs)
12"	12.750"	14.37"	17.80"	7.09"	4	¾ x 4¾"	56.1 lbs
300	(323.9 mm)	(365 mm)	(452 mm)	(180 mm)			(25.5 Kgs)
14"	14.000"	16.26"	19.25"	10.08"	4	1 x 4½"	90.6 lbs
350	(355.6 mm)	(413 mm)	(489 mm)	(256 mm)			(41.2 Kgs)
16"	16.000"	18.39"	21.30"	10.08"	4	1 x 4½"	97.2 lbs
400	(406.4 mm)	(467 mm)	(541 mm)	(256 mm)			(44.2 Kgs)
18"	18.000"	20.28"	23.43"	10.08"	4	1 x 4½"	111.1 lbs
450	(457.2 mm)	(515 mm)	(595 mm)	(256 mm)			(50.5 Kgs)
20"	20.000"	22.36"	25.63"	10.08"	4	1 x 4½"	136.0 lbs
500	(508.0 mm)	(568 mm)	(651 mm)	(256 mm)			(61.8 Kgs)

* Shurjoint recommends the use of a silicone based lubricant for the HDPE series.

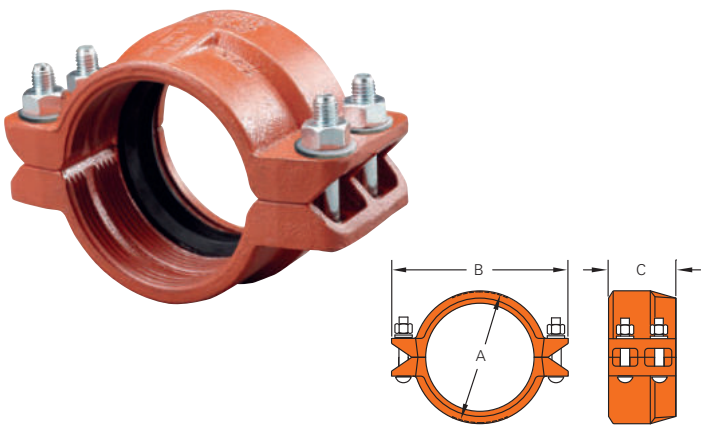
dimensions

H305 ISO size

pipe O.D.		dimensions			bolt		weight
min.	max.	A	B	C	no.	size	
50 mm	50.5 mm	72 mm	115 mm	105 mm	4	M10 x 55 mm	1.5 Kgs
63 mm	63.6 mm	85 mm	128 mm	105 mm	4	M10 x 55 mm	1.9 Kgs
75 mm	75.7 mm	97 mm	140 mm	105 mm	4	M10 x 55 mm	2.4 Kgs
90 mm	90.9 mm	113 mm	169 mm	105 mm	4	M12 x 75 mm	3.3 Kgs
110 mm	111.0 mm	139 mm	181 mm	112 mm	4	M12 x 75 mm	4.1 Kgs
140 mm	141.3 mm	169 mm	210 mm	118 mm	4	M12 x 75 mm	4.5 Kgs
160 mm	161.5 mm	190 mm	232 mm	118 mm	4	M12 x 97 mm	5.6 Kgs
180 mm	181.7 mm	211 mm	253 mm	118 mm	4	M12 x 97 mm	7.5 Kgs
200 mm	201.8 mm	236 mm	305 mm	127 mm	4	M16 x 90 mm	9.4 Kgs
225 mm	226.4 mm	261 mm	330 mm	127 mm	4	M16 x 90 mm	11.3 Kgs
250 mm	252.3 mm	289 mm	351 mm	134 mm	4	M16 x 110 mm	12.7 Kgs
280 mm	281.7 mm	319 mm	406 mm	134 mm	4	M20 x 120 mm	18.4 Kgs
315 mm	317.9 mm	354 mm	438 mm	134 mm	4	M20 x 120 mm	16.7 Kgs
355 mm	357.2 mm	412 mm	489 mm	256 mm	4	M24 x 110 mm	41.2 Kgs
400 mm	402.4 mm	462 mm	540 mm	256 mm	4	M24 x 110 mm	44.2 Kgs
450 mm	452.7 mm	515 mm	595 mm	256 mm	4	M24 x 110 mm	57.7 Kgs

* Shurjoint recommends the use of a silicone based lubricant for the HDPE series.

H307 HDPE transition coupling



The Shurjoint Model H307 transition coupling provides for a fast, easy and direct transition from HDPE pipe and or fittings to grooved end steel pipe (IPS).

dimensions
H307 IPS size



		dimensions			bolt		
nom. size	pipe O.D.	A	B	C	no.	size	weight
2"	2.375"	3.39"	5.79"	3.11"	4	½ x 2½"	4.4 lbs
50	(60.3 mm)	(86 mm)	(147 mm)	(79 mm)			(2.0 Kgs)
3"	3.500"	4.49"	6.89"	3.11"	4	½ x 3"	5.9 lbs
80	(88.9 mm)	(114 mm)	(175 mm)	(79 mm)			(2.7 Kgs)
4"	4.500"	5.75"	8.23"	3.74"	4	½ x 3"	8.4 lbs
100	(114.3 mm)	(146 mm)	(209 mm)	(95 mm)			(3.8 Kgs)
6"	6.625"	8.00"	11.02"	3.74"	4	¾ x 3½"	12.5 lbs
150	(168.3 mm)	(203 mm)	(280 mm)	(95 mm)			(5.7 Kgs)
8"	8.625"	10.51"	13.46"	4.25"	4	¾ x 3½"	21.3 lbs
200	(219.1 mm)	(267 mm)	(342 mm)	(108 mm)			(9.7 Kgs)
10"	10.750"	12.64"	16.69"	5.00"	4	¾ x 4¾"	35.2 lbs
250	(273.0 mm)	(321 mm)	(424 mm)	(127 mm)			(16.0 Kgs)
12"	12.750"	14.76"	19.02"	5.00"	4	¾ x 4¾"	43.1 lbs
300	(323.9 mm)	(375 mm)	(483 mm)	(127 mm)			(19.6 Kgs)

* Shurjoint recommends the use of a silicone based lubricant for use with the HDPE series.

dimensions
H307 IOS size

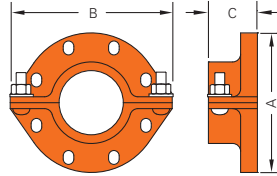


pipe O.D.		dimensions			bolt		
HDPE	ips	A	B	C	no.	size	weight
63.0 mm	60.3 mm	86 mm	146 mm	73 mm	4	M10 x 55 mm	1.5 Kgs
75.0 mm	73.0 mm	97 mm	159 mm	73 mm	4	M10 x 55 mm	1.9 Kgs
90.0 mm	88.9 mm	114 mm	178 mm	73 mm	4	M12 x 75 mm	2.3 Kgs
110.0 mm	114.3 mm	144 mm	203 mm	76 mm	4	M12 x 75 mm	2.8 Kgs
160.0 mm	165.1 mm	194 mm	254 mm	76 mm	4	M12 x 97 mm	6.2 Kgs
160.0 mm	168.3 mm	198 mm	257 mm	76 mm	4	M12 x 97 mm	6.2 Kgs
200.0 mm	219.1 mm	256 mm	321 mm	85 mm	4	M16 x 90 mm	6.4 Kgs
250.0 mm	273.0 mm	321 mm	394 mm	97 mm	4	M20 x 120 mm	11.9 Kgs
315.0 mm	323.9 mm	375 mm	451 mm	97 mm	4	M20 x 120 mm	12.7 Kgs

* Shurjoint recommends the use of a silicone based lubricant for the HDPE series.

H312 HDPE flange adapter

ANSI class 125/150



The Shurjoint Model H312 HDPE flange adapter provides for the direct transition from HDPE pipe and or fittings to ANSI Class 125 or 150 flanged components.

dimensions

H312 IPS size / ANSI class 125/150



nom. size	pipe O.D.	dimensions			draw bolt / nut		flange bolt nut*		weight
		A	B	C	no.	size	no.	size	
3"	3.500"	7.75"	8.86"	3.10"	2	5/8 x 2 1/8"	4	5/8"	10.6 lbs
80	(88.9 mm)	(197 mm)	(225 mm)	(79 mm)					(4.8 Kgs)
4"	4.500"	9.00"	10.25"	3.10"	2	5/8 x 2 1/8"	8	5/8"	15.0 lbs
100	(114.3 mm)	(229 mm)	(260 mm)	(79 mm)					(6.8 Kgs)
6"	6.625"	11.00"	12.25"	3.75"	2	5/8 x 2 1/8"	8	3/4"	21.5 lbs
150	(168.3 mm)	(279 mm)	(311 mm)	(95 mm)					(9.8 Kgs)
8"	8.625"	13.50"	14.75"	3.42"	2	3/4 x 2 3/8"	8	3/4"	28.8 lbs
200	(219.1 mm)	(343 mm)	(375 mm)	(87 mm)					(13.1 Kgs)
10"	10.750"	16.00"	21.00"	4.25"	4	3/4 x 2 3/8"	12	7/8"	42.9 lbs
250	(273.0 mm)	(406 mm)	(533 mm)	(108 mm)					(19.5 Kgs)
12"	12.750"	19.02"	24.00"	4.25"	4	3/4 x 2 3/8"	12	7/8"	51.5 lbs
300	(323.9 mm)	(483 mm)	(610 mm)	(108 mm)					(23.4 Kgs)

- * Shurjoint recommends the use of a silicone based lubricant for use with the HDPE series.
- * Flange bolts and nuts are to be supplied by installer.

dimensions

H312 ISO size / PN 10 / 16



HDPE	pipe O.D.	dimensions			draw bolt / nut		flange bolt nut*		weight
	steel	A	B	C	no.	size	no.	size	
63 mm	60.3 mm	165 mm	197 mm	79 mm	2	M16 x 54 mm	4	M16 mm	4.3 Kgs
90 mm	88.9 mm	200 mm	241 mm	79 mm	2	M16 x 54 mm	8	M16 mm	4.8 Kgs
110 mm	114.3 mm	220 mm	260 mm	79 mm	2	M16 x 54 mm	8	M16 mm	6.8 Kgs
160 mm	165.1 mm	285 mm	330 mm	82 mm	2	M20 x 60 mm	8	M20 mm	9.8 Kgs
200 mm	219.1 mm	340 mm	400 mm	108 mm	2	M20 x 60 mm	12 (PN 16)	M20 mm	13.1 Kgs
250 mm	273.0 mm	405 mm	533 mm	108 mm	4	M24 x 100 mm	12	M24 mm	19.5 Kgs
315 mm	323.9 mm	460 mm	587 mm	108 mm	4	M24 x 100 mm	12	M24 mm	28.5 Kgs

- * Shurjoint recommends the use of a silicone based lubricant for use with the HDPE series.
- * Flange bolts and nuts are to be supplied by installer.